Capstone Project Document: Planeswalker's Primer

Introduction

Purpose

What is the problem or the opportunity that the project is investigating?

Magic: The Gathering has a notoriously steep learning curve, with new players often overwhelmed by complex rules, terminology, and strategic concepts. The opportunity lies in creating a centralised, beginner-friendly digital reference platform that provides comprehensive information about MTG fundamentals in an accessible format.

Why is this problem valuable to address?

- MTG has over 35 million players worldwide, but player retention among beginners remains a significant challenge
- Local game stores struggle to onboard new players effectively without dedicated reference resources
- The game's complexity creates a barrier to entry that prevents market growth
- Existing resources are fragmented across wikis, rulebooks, and community sites

What is the current state?

- New players frequently abandon the game due to feeling lost or intimidated by complexity
- Learning resources are scattered across multiple platforms with varying quality
- No single comprehensive platform designed specifically for beginners
- Local game stores lack standardised reference materials

What is the desired state?

- A single, comprehensive reference platform that organises MTG fundamentals
- Easy-to-navigate information hub for quick lookups during gameplay
- Interactive tools for card discovery and reference
- Mobile-friendly access for learning during gameplay

Has this problem been addressed by other projects?

Yes, several projects exist but with different focuses:

- MTG Wiki: Comprehensive but overwhelming for beginners
- EDHRec: Focused on deck building for experienced players
- Official MTG website: Marketing-focused rather than education-focused
- YouTube tutorials: Unstructured and difficult to navigate systematically

Industry/Domain

What is the industry/domain?

The trading card game (TCG) industry, specifically focused on Magic: The Gathering - the world's first and most successful TCG with over 30 years of history.

What is the current state of this industry?

- Market size: Global TCG market valued at £9.1 billion in 2023
- Digital transformation: Shift towards digital companions for physical games
- Challenges: Competition from digital-only card games, complexity barriers for new players
- Growth: Consistent 20% year-over-year growth in the MTG segment

What is the overall industry value-chain?

1. **Game Publisher** (Wizards of the Coast) → Designs and produces cards
2. **Distributors** → Supply products to retailers
3. **Local Game Stores** → Sell products and host events
4. **Players** → Purchase and play the game
5. **Secondary Market** → Card trading and resale
6. **Content Creators** → Tutorials, gameplay videos, strategy guides

What are the key concepts in the industry?

- **Mana System**: Resource management using five colours
- **Card Types**: Creatures, Spells, Artefacts, Enchantments, Planeswalkers, Lands
- **Game Phases**: Untap, Upkeep, Draw, Main, Combat, End
- **Stack and Priority**: Timing and interaction rules
- **Formats**: Different ways to play (Standard, Modern, Commander, etc.)
- **Deck Construction**: Building 60+ card decks with strategic synergies

Is the project relevant to other industries?

Yes, the organised reference approach applies to:

- Other complex tabletop games
- Educational technology platforms
- Digital reference systems
- Community onboarding tools

Stakeholders

Who are the stakeholders?

Primary Stakeholders:

- **New MTG Players**: Ages 13-35 seeking to understand game fundamentals
- **Returning Players**: Former players needing quick reference on rules and mechanics

Secondary Stakeholders:

- **Local Game Store Owners**: Need reference tools to support new player onboarding
- **Experienced Players**: Mentoring newcomers and needing teaching resources
- **Parents**: Helping children learn the game safely

Tertiary Stakeholders:

- **Content Creators**: Reference material for tutorial creation
- **Tournament Organisers**: Ensuring new players understand competitive rules

Why do they care about this software?

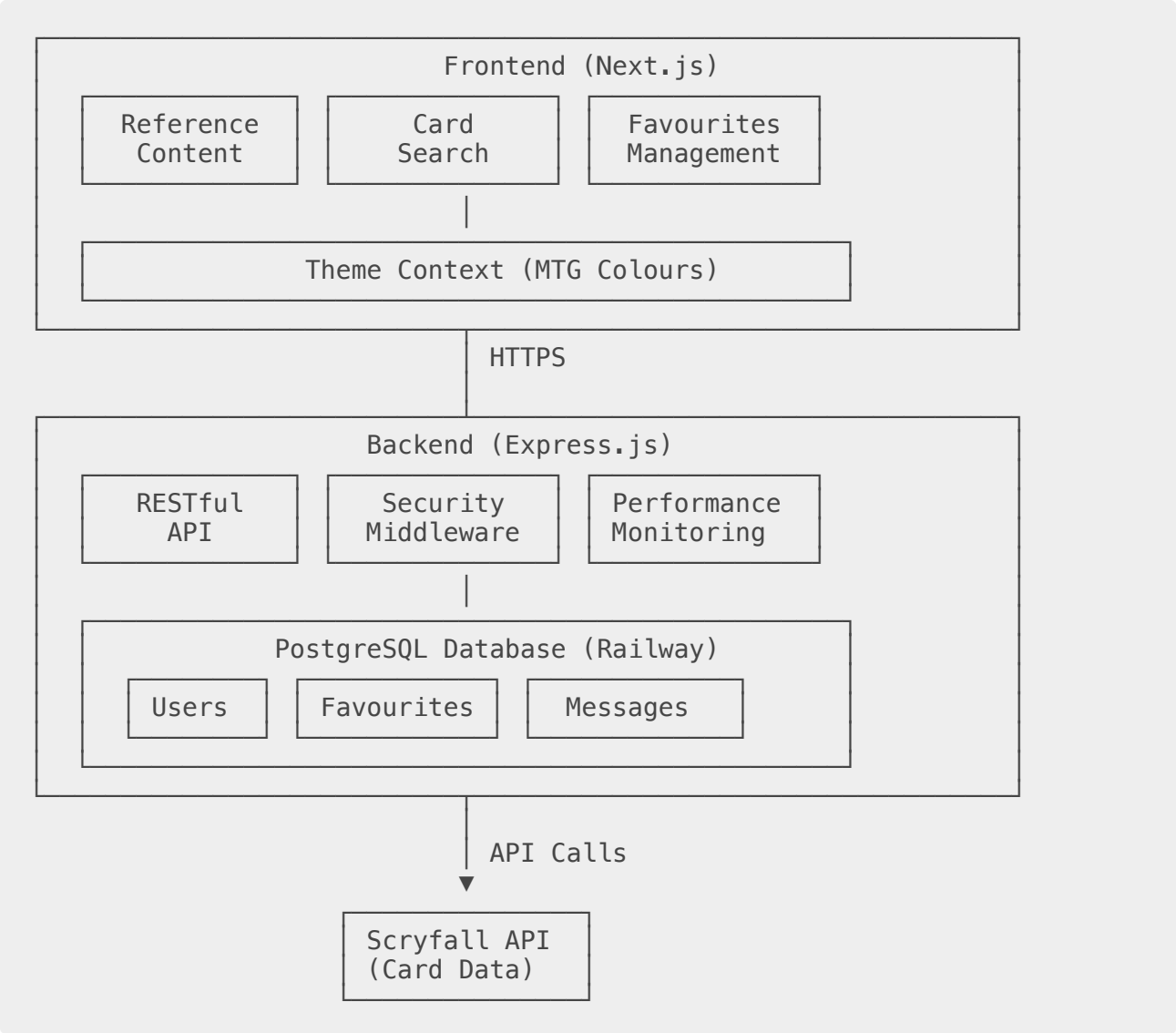
- **New Players**: Quick access to comprehensive game information
- **Store Owners**: Improves customer experience and retention
- **Experienced Players**: Reliable reference for teaching
- **Parents**: Provides safe, organised information resource

What are the stakeholders' expectations?

- Clear, accurate information following official rules
- Easy navigation to find specific information quickly
- Mobile-friendly interface for reference during games
- Regular updates as new mechanics are introduced
- Free access to core reference materials
- Dark theme for comfortable extended reading

Product Description

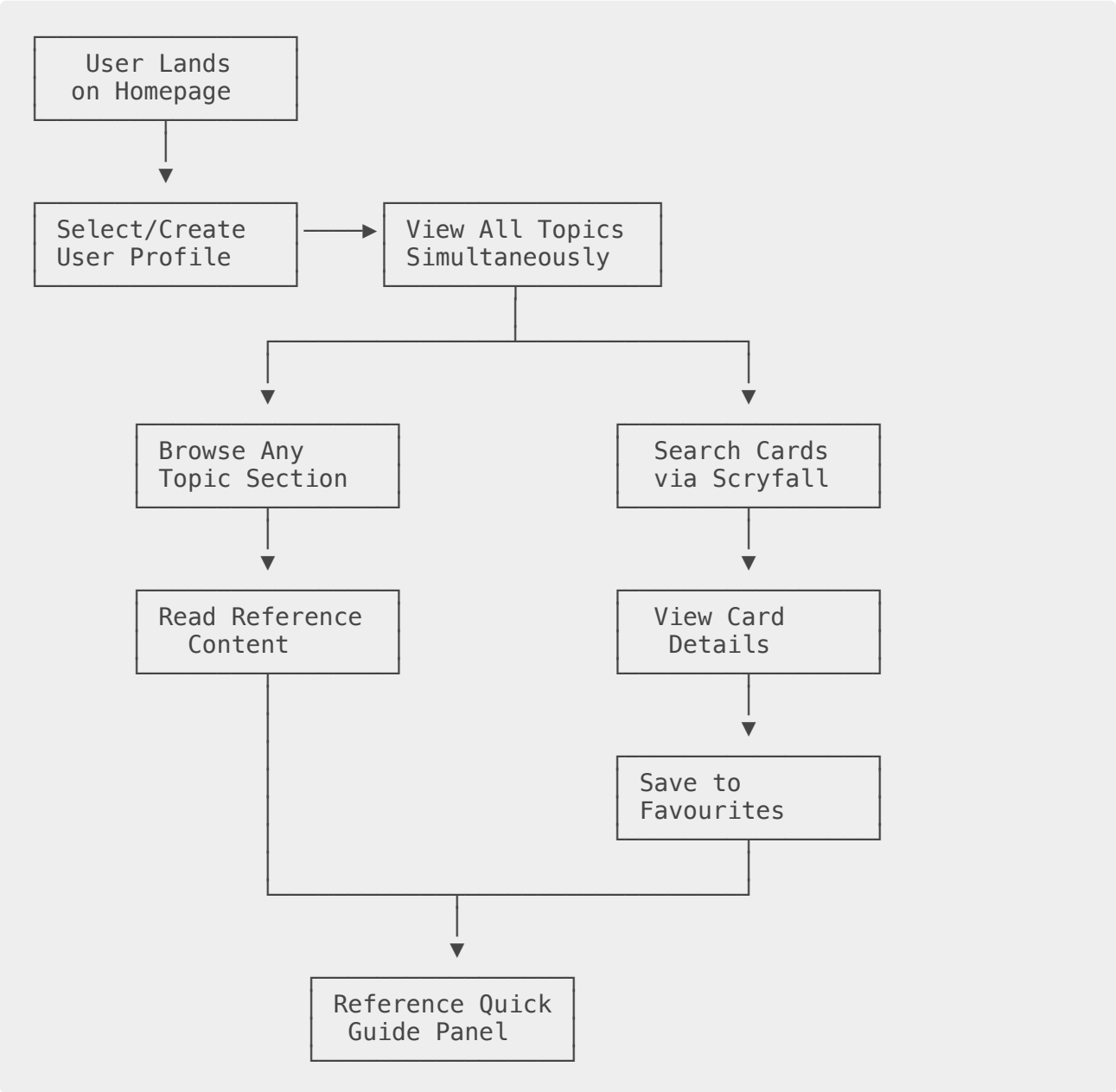
Architecture Diagram



User Stories

#	User Story	Description	Priority	Status
1	Access MTG Reference	As a new player, I want to access comprehensive MTG information in one place so that I can look up rules quickly	High	✅ Complete
2	Search MTG Cards	As a user, I want to search for cards and view detailed information so that I can learn about specific cards	High	✅ Complete
3	Save Favourites	As a user, I want to save my favourite cards for quick access so that I can reference them later	Medium	✅ Complete
4	Browse Mechanics	As a player, I want to explore MTG mechanics with detailed explanations so that I understand how abilities work	High	✅ Complete
6	Mobile Access	As a player, I want to access the platform on my mobile device so that I can reference information during games	High	✅ Complete

User Flow


User Lands on Homepage
Select/Create User Profile
View All Topics Simultaneously
Browse Any Topic Section
Search Cards via Scryfall
Read Reference Content
View Card Details
Save to Favourites
Reference Quick Guide Panel

Wireframe Design

The application features a mobile-first dark theme design optimised for gameplay environments. Key design elements include:

- **Dark Theme**: Reduces eye strain during extended reading sessions
- **MTG Colour Integration**: UI elements themed around the five MTG colours
- **Responsive Layout**: Seamless experience across mobile, tablet, and desktop
- **Card-Based Components**: Information presented in familiar card-like containers
- **High Contrast**: Ensures readability in various lighting conditions

Interactive Figma Prototype: View Mobile Design

Open Questions/Out of Scope

Features considered out of scope:

- Structured learning paths with progression tracking
- Curriculum-based lesson sequences
- Prerequisite systems for content
- Learning progress tracking
- Advanced strategy guides and meta-game analysis
- Deck building tools and recommendations
- Multiplayer features or social networking
- Tournament tracking and results
- Real-time gameplay simulation
- Password authentication for users
- Commercialisation of the application
- CI/CD pipeline implementation
- Automated testing workflows
- GitHub Actions integration
- Continuous deployment processes

Non-functional Requirements

Security Requirements

- **Security Headers**: Robust implementation including CSP, HSTS, and X-Frame-Options
- **Rate Limiting**: 1000 requests per 15 minutes to prevent API abuse
- **Input Validation**: All user inputs sanitised and validated
- **CORS Configuration**: Secure cross-origin request handling
- **Error Handling**: Secure error messages without information leakage
- **HTTPS Enforcement**: All communications encrypted in transit

Performance Requirements

- **Response Time**: API endpoints respond within 500ms
- **Caching**: 5-minute cache for static content
- **Compression**: Middleware reduces payload sizes
- **Database Pooling**: Efficient connection management
- **Load Handling**: Support 100 concurrent users
- **Mobile Performance**: Optimised for 3G connections

Usability Requirements

- **Mobile-First Design**: Full functionality on devices 320px and wider
- **Accessibility**: Keyboard navigation and screen reader support
- **Learning Curve**: New users can navigate content within 5 minutes
- **Error Recovery**: Clear error messages with suggested actions
- **Offline Capability**: Basic content accessible without connection

Reliability Requirements

- **Availability**: 99.5% uptime target
- **Data Integrity**: Transactional consistency for user data
- **Graceful Degradation**: Core features work if external APIs fail
- **Backup Strategy**: Daily database backups
- **Recovery Time**: < 1 hour for critical failures

Standards Compliance

- **API Design**: RESTful following OpenAPI 3.0 specification
- **Code Standards**: ESLint configuration for consistency
- **Web Standards**: HTML5, CSS3, ES6+ JavaScript
- **Accessibility**: WCAG 2.1 Level AA compliance target
- **Documentation**: JSDoc comments for all major functions

Project Planning

Development Timeline

Phase 1: Foundation (Weeks 1-2)

- ✅ Project setup and repository initialisation
- ✅ Database schema design and implementation
- ✅ Basic frontend and backend architecture
- ✅ Development environment configuration

Phase 2: Core Features (Weeks 3-5)

- ✅ MTG reference content organisation
- ✅ User management system
- ✅ Favourites functionality
- ✅ Dark theme implementation

Phase 3: Integration (Weeks 6-7)

- ✅ Scryfall API integration
- ✅ Performance monitoring
- ✅ Security implementation
- ✅ Theme system with MTG colours

Phase 4: Polish & Deployment (Weeks 8-9)

- ✅ Production deployment
- ✅ Testing suite completion
- ✅ API documentation
- ✅ Project documentation

Risk Management

- **Technical Risks**: Mitigated through extensive testing and fallback strategies
- **API Dependency**: Cached responses and graceful degradation if Scryfall unavailable
- **Scope Creep**: Clear out-of-scope definition and focused feature set
- **Performance**: Monitoring and optimisation throughout development

Testing Strategy

Quality Assurance Approach

- **Test Coverage**: Extensive coverage across frontend and backend components
- **Testing Framework**: Jest for both React components and API endpoints
- **Manual Testing**: User experience validation with family and friends
- **Security Testing**: Rate limiting and input validation verification
- **Performance Testing**: Load testing and response time monitoring

Feature Testing

- **Reference Content**: Information accuracy and navigation flow
- **Card Search**: API integration and error handling
- **Favourites System**: CRUD operations and data persistence
- **User Management**: Profile creation and switching
- **Theme System**: Colour switching and persistence

Edge Case Handling

- **Network Failures**: Graceful degradation with helpful error messages
- **Invalid Input**: Comprehensive validation with user guidance
- **Rate Limiting**: Clear feedback when limits exceeded
- **Database Errors**: Fallback behaviour and error recovery
- **Missing Data**: Default values and placeholder content

Testing Process

Tests are run manually during development:

- Backend: `cd backend && npm test`
- Frontend: `cd frontend && npm test`

Implementation

Technology Stack

- **Frontend**: Next.js 13+, React 18, CSS-in-JS with custom global styles
- **Backend**: Node.js with Express.js, Helmet security middleware
- **Database**: PostgreSQL with connection pooling
- **External APIs**: Scryfall API for MTG card data
- **Testing**: Jest framework with unit and integration tests
- **Deployment**: Vercel (frontend) and Railway (backend)

Deployment Considerations

- **Environment Management**: Separate development and production configurations
- **Security**: HTTPS enforcement and security headers
- **Monitoring**: Real-time performance tracking
- **Scalability**: Stateless architecture ready for horizontal scaling
- **Manual Deployment Process**: Controlled releases through platform dashboards

Manual Deployment Process

1. **Frontend Deployment**: Push to GitHub, deploy through Vercel dashboard
2. **Backend Deployment**: Push to GitHub, deploy through Railway dashboard
3. **Database Updates**: Execute migrations via Railway PostgreSQL console
4. **Verification**: Test both staging and production environments

End-to-end Solution

Objectives Achievement

The software successfully meets its objectives of providing a comprehensive, beginner-friendly MTG reference platform:

1. **Comprehensive Reference Coverage**: ✅ All fundamental concepts covered
2. **User-Friendly Interface**: ✅ Intuitive navigation and dark theme
3. **Mobile Accessibility**: ✅ Fully responsive design
4. **Card Reference System**: ✅ Integrated Scryfall search
5. **Personal Reference Tools**: ✅ Favourites system implemented
6. **Performance**: ✅ Sub-500ms response times achieved
7. **Security**: ✅ Robust security measures implemented

User Feedback

The application was extensively tested among family and friends, with positive feedback on:

- Easy access to comprehensive information
- Clear organisation of game mechanics
- Mobile usability during actual games
- Quick reference capabilities

Production Metrics

- **Deployment Success**: Live on production environments
- **Performance**: Meets all performance targets
- **Reliability**: No critical failures in production
- **User Adoption**: Successful onboarding of test users

References

Code Repository

- **GitHub**: github.com/tatoslover/Capstone

External Resources

- **Scryfall API**: Complete MTG card database and high-resolution images
- **MTG Wiki**: Authoritative source for game mechanics and rules
- **Wizards of the Coast**: Official MTG comprehensive rules (reference only)

Technology Stack

- **Next.js**: React framework for production applications
- **Express.js**: Fast, unopinionated web framework for Node.js
- **PostgreSQL**: Advanced open-source relational database
- **Jest**: JavaScript testing framework
- **Vercel**: Frontend deployment platform
- **Railway**: Backend deployment and database hosting
- **Helmet**: Security middleware for Express applications
- **Swagger/OpenAPI**: API documentation framework

Libraries and Dependencies

- **Frontend**: React 18, Next.js 13, CSS-in-JS
- **Backend**: Express 4, pg (PostgreSQL client), cors, helmet
- **Testing**: Jest, React Testing Library
- **Documentation**: Swagger UI Express, OpenAPI 3.0
- **Performance**: Compression middleware, response time tracking